

# The EUV Snapshot Imaging Spectrograph

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## 1. INTRODUCTION

The light emitted by the solar transition region (TR) and corona varies significantly as a function of position, wavelength, and time. When viewed from Earth, the spectral radiance from the Sun can be written as:  $I(x, y, \lambda, t)$ , where  $x$  and  $y$  are the helioprojective Cartesian coordinates (W. T. Thompson 2006),  $\lambda$  is wavelength, and  $t$  is time. The ideal solar imaging spectrograph would capture  $I(x, y, \lambda, t)$  with high resolution in  $x$ ,  $y$ ,  $\lambda$ , and  $t$  and over a wide field of view (FOV), wavelength range, and time period. Of course, the temporal dimension is privileged, so we often reduce the problem to capturing a 3D spatial/spectral cube at a particular time  $t_0$ :  $I(x, y, \lambda, t_0)$ . Since we use 2D detectors, this means that we must find a way to flatten the 3D cube into two dimensions without losing information.

One obvious way to accomplish this is to multiplex one of the three remaining dimensions in time. Narrowband, tunable filters, such as the GREGOR Fabry–Pérot Interferometer (K. G. Puschmann et al. 2012) or CRISP at the Swedish Solar Telescope (G. B. Scharmer et al. 2008), multiplex the wavelength dimension in time, and can change the selected wavelength in 50 ms or less (G. B. Scharmer et al. 2008), but the technology does not exist to use this technique for wavelengths shorter than  $\sim 150$  nm (J.-P. Wuelser et al. 2000). The nearest extreme ultraviolet (EUV) equivalent is a multilayer-coated imager such as the Transition Region and Coronal Explorer (TRACE) (B. N. Handy et al. 1999) or the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), which abandons the wavelength dimension entirely and integrates over a passband holding many emission lines. Two such passbands can be compared to infer a Doppler shift (T. Sakao et al. 1999), but weaker

lines within them bound the velocity resolution near  $\sim 1000 \text{ km s}^{-1}$  (K. Kobayashi et al. 2000), coarser than the flows that characterize TR dynamics.

A spatial dimension can be multiplexed instead. An entrance slit admits a single column of the scene, so the detector records  $\lambda$  against  $y$  without ambiguity, and  $x$  is recovered by stepping the slit across the target. The Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014) and the Spectral Imaging of the Coronal Environment (SPICE) aboard Solar Orbiter (SPICE Consortium et al. 2020) work this way, and the spectra they return are the most faithful of any technique discussed here. What is sacrificed is simultaneity: neighboring columns of the reconstructed cube are separated in time by the raster, and structure that evolves faster than the raster completes is smeared along  $x$ . The Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020, 2022; M. C. M. Cheung et al. 2022) narrows this gap considerably by ruling 37 slits across the field and separating their overlapping spectra afterward, which shortens the raster of an active region to as little as 12 s.

A third strategy divides the detector rather than the exposure. An integral field spectrograph slices the field of view and reformats those slices so that each is dispersed onto its own region of the detector, so the entire cube arrives in one exposure, at the cost of trading field of view against wavelength range. The microlensed hyperspectral imager of M. van Noort et al. (2022) demonstrates the technique at visible wavelengths, and the Solar eruption Integral Field Spectrograph (SNIFS) (V. L. Herde et al. 2024), flown in 2025, demonstrates it in the far ultraviolet (FUV). Extending it to the EUV remains out of reach of available optics.

The last strategy is to allow the flattening to be ambiguous and to undo it afterward. A spectrograph with no entrance slit disperses the whole field at once, so every emission line forms an image of the Sun and

those images superimpose on the detector. The resulting frame, an *overlappogram*, encodes  $x$  and  $\lambda$  along a single axis, and the two cannot be separated within one exposure. Overlappograms are as old as spaceborne solar spectroscopy, having been recorded by the Naval Research Laboratory’s (NRL’s) S082A spectroheliograph (R. Tousey et al. 1973, 1977), which yielded both a census of EUV transitions (U. Feldman et al. 1985) and, much later, flare line ratios (F. P. Keenan et al. 2006); the technique is in use today at soft X-ray wavelengths by the Marshall Grazing Incidence X-ray Spectrometer (MaGIXS) (S. L. Savage et al. 2023). What has changed since S082A is that the ambiguity has become tractable. Reconstructing the cube from superimposed images is a tomographic problem (A. C. Kak & M. Slaney 1988), in which each diffraction order views the cube from a different angle and the number of independent views limits how much of the line profile can be recovered (M. R. Descour et al. 1997). Computed tomography imaging spectrographs (CTISs) (T. Okamoto & I. Yamaguchi 1991; F. V. Bulygin et al. 1991; M. Descour & E. Dereniak 1995) carry this to its extreme, projecting as many as 25 orders onto a single detector, with computational cost rising accordingly (N. Hagen & E. L. Dereniak 2008; N. Hagen & M. W. Kudenov 2013). Fewer views can suffice when the scene is sparse: C. DeForest et al. (2004) synthesized a magnetogram from a single exposure using only two dispersed orders. Inversion methods for overlapping spectral images have advanced considerably since (A. R. Winebarger et al. 2019; J. M. Davila et al. 2019; U. Kamaci & F. Kamalabadi 2026).

The Multi-Order Solar EUV Spectrograph (MOSES) (J. L. Fox et al. 2010; J. L. Fox 2011) brought this strategy to the solar EUV. A single concave grating forms three images at once, the undispersed  $m = 0$  order and the  $m = \pm 1$  orders, on three detectors, while a multilayer coating narrows the passband to a few emission lines so that the cube to be recovered is sparse enough to invert. Two flights showed that the concept works. J. L. Fox et al. (2010) measured the Doppler shift and width of explosive event line profiles in the TR; T. Rust & C. C. Kankelborg (2019) resolved bimodal profiles in quiet Sun events; H. T. Courrier & C. C. Kankelborg (2018) recovered Doppler maps using local correlation tracking; and J. D. Parker & C. C. Kankelborg (2022) separated the contributions of individual spectral lines. The same flights exposed where the design binds. Only three views are available, one of which is undispersed and therefore contributes no spectral information, which caps the number of degrees of freedom that an inversion can recover; and because the dispersed orders lie in a

plane, they fill the cylindrical volume of a sounding rocket payload poorly.

The EUV Snapshot Imaging Spectrograph (ESIS) was designed to relax both constraints. An array of gratings arranged about the optical axis re-images the prime focus of a single primary mirror, each dispersing at a different angle, so that every detector records a dispersed view and the views are distributed around the payload rather than confined to a plane. ESIS flew for the first time on 2019 September 30, and first results from that flight are reported by J. D. Parker et al. (2022). An earlier form of the design was described by H. T. Courrier (2020); the instrument that flew differs from it, most consequentially in the removal of the primary mask, which admits vignetting into the system. This paper describes the instrument as it was built and flown. In Section 2 we explain how experience with MOSES shaped the design of ESIS. Section 3 states the scientific objectives and the requirements they impose. Section 4 describes the instrument itself, and Section ?? the mission profile.

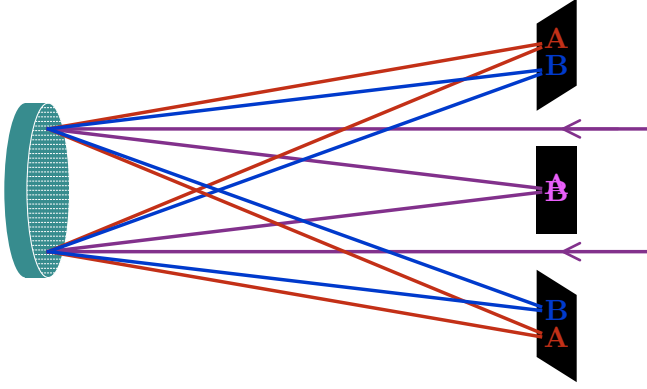
## 2. THE ESIS CONCEPT

A primary goal of the ESIS instrument is to improve upon the imaging spectroscopy demonstrated by MOSES. Therefore, the design of the new instrument draws heavily from experiences and lessons learned through two flights of MOSES. ESIS and MOSES are both CTIS instruments. As such, both produce overlappograms of a narrow portion of the solar spectrum, with the goal of enabling the reconstruction of a spectral line profile at every point in the field of view. The similarities end there, however, as the optical layout of ESIS differs significantly from that of MOSES. In this section, we detail some difficulties and limitations encountered with MOSES, then describe how the new design of ESIS addresses these issues.

### 2.1. Limitations of the MOSES Design

The MOSES design features a single concave diffraction grating forming images on three charge-coupled device (CCD) detectors (J. L. Fox et al. 2010) (Figure 1). The optical path is folded in half by a single flat secondary mirror (omitted in Figure 1). Provided that the three cameras are positioned correctly, this arrangement allows the entire telescope to be brought into focus using only the central (undispersed) order and a visible light source. Unfortunately this design uses volume inefficiently for two reasons. First, the lack of magnification by the secondary mirror limits the folded length of the entire telescope to be no less than half of the 5 m focal length of the grating (J. L. Fox et al. 2010; J. L. Fox 2011). Second, the resolving power of the instrument is limited by the ratio of camera separation

to pixel size. To achieve the maximum dispersion of  $29 \text{ km pix}^{-1} \text{ s}^{-1}$  (J. L. Fox et al. 2010), the outboard orders are imaged as far apart as possible in the  $\sim 0.56 \text{ m}$  diameter envelope of the rocket payload. This planar arrangement leaves much unused space in the cylindrical volume of the payload.



**Figure 1.** Schematic diagram of the MOSES instrument. Incident light from the right forms an undispersed image on the central  $m = 0$  CCD. Dispersed images are formed on the outboard  $m = \pm 1$  CCDs. The imaging of longer and shorter wavelengths is indicated by the use of red and blue letters respectively.

Furthermore, the monolithic secondary, though it confers the focus advantage noted above, does not allow efficient placement of the  $m = \pm 1$  CCDs. For all practical purposes, the payload can only accommodate three diffraction orders ( $m = -1, 0, +1$ ). Therefore, *MOSES can only collect, at most, three pieces of information at each point in the field of view.* From this, it is not reasonable to expect the reconstruction of more than three degrees of freedom for each spectral line, except in the case of very compact, isolated features such as those described by J. L. Fox et al. (2010) and T. Rust & C. C. Kankelborg (2019). Consequently, it is a reasonable approximation to say that MOSES is sensitive primarily to spectral line intensities, shifts, and widths (C. C. Kankelborg & R. J. Thomas 2001). With any tomographic apparatus, the degree of detail that can be resolved in the object depends critically on the number of viewing angles (A. C. Kak & M. Slaney 1988; M. R. Descour et al. 1997; N. Hagen & E. L. Dereniak 2008). So it is with the spectrum we observe with MOSES: more dispersed images are required to confer sensitivity to finer spectral details such as higher moments of the spectral line shape.

A related issue stems from the use of a single dispersion plane. Since the solar corona and transition region are structured by magnetic fields, the scene tends to be dominated by field-aligned structures such as

loops (R. Rosner et al. 1978; R. M. Bonnet et al. 1980). When the MOSES dispersion direction happens to be aligned nearly perpendicular to the magnetic field, filamentary structures on the transition region serve almost as spectrograph slits unto themselves. The estimation of Doppler shifts then becomes a simple act of triangulation, and broadenings are also readily diagnosed (J. L. Fox et al. 2010; H. T. Courrier & C. C. Kankelborg 2018). A double-peaked profile can also be observed with sufficiently isolated features (T. Rust & C. C. Kankelborg 2019). Unfortunately, solar magnetic fields in the transition region are quite complex and do not have a global preferred direction. In cases where the field is nearly parallel to the instrument dispersion, spectral shifts and broadenings are not readily apparent.

The single diffraction grating also leads to a compromise in the optical performance of the instrument. Since the MOSES grating forms images in three orders simultaneously, there are not enough degrees of freedom in the optical system to control aberrations in all three spectral orders. During the first mission, MOSES was flown with a small amount of defocus (T. Rust & C. C. Kankelborg 2019), which exacerbated the inter-order point-spread function (PSF) variation and caused the individual PSFs to span several pixels (T. Rust & C. C. Kankelborg 2019; S. Atwood & C. Kankelborg 2018). The outboard PSFs were elongated, with their major axes at different angles. This resulted in spurious spectral features that require additional consideration (S. Atwood & C. Kankelborg 2018) and further increase the complexity of the inversion process (T. Rust & C. C. Kankelborg 2019; H. T. Courrier & C. C. Kankelborg 2018).

As MOSES lacks a field stop, it is possible for the grating to image off-band radiation on the detector from solar features that lie outside the intended FOV. Although the intensity of this off-band radiation is relatively low, it is not negligible. J. D. Parker & C. C. Kankelborg (2022) compared synthetic MOSES images to the real data and found that approximately ten percent of the intensity in the zeroth order image originated from more than ten dim lines in the MOSES passband, most of them too faint to be visible in the dispersed images. This study revealed that an undispersed ( $m = 0$ ) channel, although attractive due to its lack of spatial-spectral ambiguity, is of limited utility due to spectral contamination. Moreover, the FOV should be clearly defined, and the same, for each image, so that the spectral and spatial content of each image is unambiguous.

Finally, the exposure cadence of MOSES is hindered by a  $\sim 6 \text{ s}$  readout time for the CCDs (J. L. Fox 2011). The observing interval for a solar sounding rocket flight is typically about five minutes. Consequently, every sec-

ond of observing time is precious, both to achieve adequate exposure time and to catch the full development of dynamical phenomena. The MOSES observing duty cycle is  $\sim 50\%$  since it is limited by the readout time of its CCDs. Thus, valuable observing time is lost. The readout data gap compelled us to develop a sequence with exposures ranging from 0.25-24 s, a careful trade-off between deep and fast exposures.

In summary, our experience leads us to conclude that the MOSES design has the following primary limitations:

1. inefficient use of volume
2. dispersion constrained by payload dimensions
3. too few dispersed images (orders)
4. single dispersion plane
5. lack of aberration control
6. poorly-defined and wavelength dependent FOV
7. spectral contamination
8. low duty cycle

In designing ESIS, we have sought to improve upon each of these points.

## 2.2. ESIS Features

The layout of ESIS (Figure 2) is a modified form of Gregorian telescope. Incoming light is brought to focus at an octagonal field stop by a parabolic primary mirror. In the ESIS layout, the secondary mirror of a typical Gregorian telescope is replaced by a segmented, octagonal array of diffraction gratings. From the field stop, the gratings re-image to CCD detectors arranged radially around the primary mirror. The gratings are blazed for first order, so that each CCD is fed by a single corresponding grating, and all the gratings are identical in design. The features of this new layout address all of the limitations described in Section 2.1, and are summarized here.

Replacing the secondary mirror with an array of concave diffraction gratings confers several advantages to ESIS over MOSES. First, the concavity of the gratings creates magnification in the ESIS optical system, which results in a shorter axial length than MOSES, without sacrificing spatial or spectral resolution. Second, the magnification and tilt of an individual grating controls the position of the dispersed image with respect to the optical axis, so that the spectral resolution is not constrained by the payload dimensions. Third, the radial symmetry of the design places the cameras closer

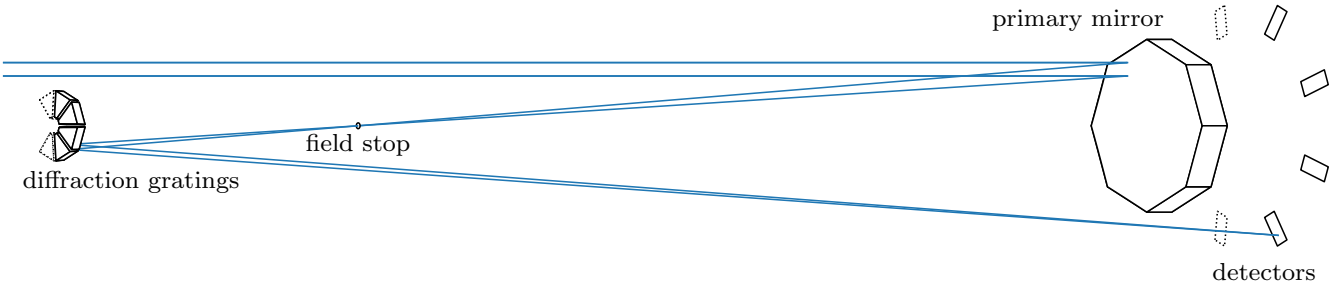
together, resulting in a more compact instrument. Furthermore, by arranging the detectors around the optical axis, more dispersed grating orders can be populated; up to eight gratings can be arrayed around the ESIS primary mirror (up to six with the current optical table). This contrasts with the three image orders available in the planar symmetry of MOSES. Taken together, these three design features make ESIS more compact than MOSES (§ 2.1 item 1), improve spectral resolution (item 2) and allow the collection of more projections to better constrain the interpretation of the data (item 3).

The ESIS gratings are arranged in a segmented array, clocked in  $45^\circ$  increments, so that there are four distinct dispersion planes. This greatly aids in reconstructing spectral line profiles since the dispersion space of ESIS occupies a 3D volume rather than a 2D plane as with MOSES. For ESIS, there is always a dispersion plane within  $22.5^\circ$  of the normal to any loop-like feature in the solar atmosphere. As discussed in Section 2.1, a nearly perpendicular dispersion plane allows a filamentary structure to serve like a spectrographic slit, resulting in a clear presentation of the spectrum. This feature addresses § 2.1 item 4.

Rather than forming images at three spectral orders from a single grating, each ESIS imaging channel has a dedicated grating. Aberrations are controlled by optimizing the grating design to form images in first order, over a narrow range of ray deviation angles. This design controls aberration well enough to allow pixel-limited imaging, avoiding the PSF mismatch problems inherent to the MOSES design (§ 2.1 item 5). In its flight configuration with gratings optimized around a  $629.732 \text{ \AA}$  wavelength, the instrument cannot be aligned and focused in visible light like MOSES. Visible gratings and a special alignment transfer procedure (§ ??) are used to align and focus ESIS.

The ESIS design also includes an octagonal field stop placed at prime focus. This confers two advantages. First, the field stop fully defines the instrument FOV, so that ESIS is not susceptible to the spectral confusion observed in MOSES data (§ 2.1 item 6). Second, each spectral line image observed by ESIS is bordered by the outline of the field stop (e.g. § ??). This aids the inversion process since outside of this sharp edge the intensity is zero for any look angle through an ESIS data cube. The size and octagonal shape of the field stop are defined by the requirement that all CCDs must see the entire FOV from edge to edge, while leaving a small margin for alignment.

Lastly, in contrast to MOSES, ESIS employs frame transfer CCDs to make optimum use of our five minutes of observing time. The ESIS design is shutterless, so



**Figure 2.** The ESIS optical layout. Dotted lines indicate the positions of unpopulated channels. The blue lines represent the path of O V through the system. The ESIS instrument is a pseudo-Gregorian design. The secondary mirror is replaced by a segmented array of concave diffraction gratings. The field stop at prime focus defines instrument spatial/spectral FOV. CCDs are arrayed around the primary mirror, each associated with a particular grating. Eight grating positions are available in principle; only six fit within the volume of the rocket payload. Four channels are populated for the first flight.

that each detector is always integrating. The result is a 100% duty cycle. The lack of downtime for readout also allows ESIS to operate at a fixed, rapid cadence of 10 s. Longer integration times can be achieved for faint features by exposure stacking (§ 2.1 item 8).

In summary, the ESIS concept addresses all the limitations of the MOSES design enumerated in § 2.1. The volume of the ESIS optical layout is smaller than MOSES by almost a factor of two, yet with a smaller PSF, improved spectral resolution, and faster exposure cadence. ESIS offers several features to improve the recovery of spectral information, including more channels, crossed dispersion planes, and a field stop.

### 3. SCIENCE OBJECTIVES

Early flights of MOSES demonstrated a working concept of simultaneous EUV imaging and spectroscopy. This concept adds a unique capability to the science that we can obtain from the EUV solar atmosphere. ESIS as designed improves upon the MOSES concept, as discussed in the previous section, and therefore improves our ability to accomplish our scientific objectives. In this section, we set forth the specific scientific objectives of the ESIS mission. It is from these objectives that we derived the quantitative science requirements (§ 3.3) that drove the ESIS design.

The ESIS mission was designed to achieve the following two overarching science goals: (1) observe magnetic reconnection in the TR, and (2) map the transfer of energy through the TR with emphasis on magnetohydrodynamic (MHD) waves. These objectives have significant overlap with the missions of IRIS (B. De Pontieu et al. 2014), the EUV Imaging Spectrograph (EIS) (J. L. Culhane et al. 2007) aboard Hinode, the EUV Normal-incidence Spectrometer (EUNIS) (J. W. Brosius et al. 2007, 2014), and a long history of FUV and EUV slit spectrographs. The ESIS instrument, however, can obtain both spatial and spectral information

co-temporally. This allows us to resolve complicated morphologies of compact TR reconnection events (as was done with MOSES (J. L. Fox 2011; T. L. Rust 2017; H. T. Courrier & C. C. Kankelborg 2018)) and observe signatures of MHD waves over a large portion of the solar disk. Therefore, in support of goal 1, we will use ESIS to map flows as a function of time and space in multiple TR reconnection events. To achieve goal 2, we will cross-correlate the evolution at multiple temperatures in the TR to map the vertical transport of energy over a wide FOV.

#### 3.1. Magnetic Reconnection Events

Magnetic reconnection describes the re-arrangement of the magnetic topology wherein magnetic energy is converted to kinetic energy resulting in the acceleration of plasma particles. Reconnection is implicated in many dynamic, high-energy solar events. Solar flares are a well-studied example (e.g. E. R. Priest & T. G. Forbes (2002) and the references therein); however, we have little hope of pointing in the right place at the right time to observe a significant flare event in a rocket flight lasting only five minutes. Instead, we will search for signatures of magnetic reconnection in TR spectral lines. A particular signature of reconnection in the TR is the explosive energy release by ubiquitous, small-scale events. These explosive events (EEs) are characterized as spatially compact ( $\approx 1.5$  Mm length (K. P. Dere 1994)) line broadenings on the order of  $100 \text{ km s}^{-1}$  (K. P. Dere et al. 1991). They are observed across a range of TR emission lines that span temperatures of 20 000–250 000 K (C II–O V) (D. Moses et al. 1994). The typical lifetime of an EE is 60–90 s (D. Moses et al. 1994; K. P. Dere 1994; K. P. Dere et al. 1991). Due to their location near quiet Sun magnetic network elements, and the presence of supersonic flows near the Alfvén speed, K. P. Dere et al. (1991) first suggested that EEs may result from

the model of fast Petschek (H. E. Petschek 1964) reconnection.

The spectral line profile of EEs may indicate the type of reconnection that is occurring in the TR (e.g. T. L. Rust (2017)). For example, the Petschek model of reconnection predicts a ‘bi-directional jet’ line profile with highly Doppler-shifted wings, but little emission from the line core (D. E. Innes & G. Tóth 1999). D. E. Innes et al. (2015) developed a reconnection model resulting from a plasmoid instability (A. Bhattacharjee et al. 2009). In contrast to the bi-directional jet, this modeled line profile has bright core emission and broad wings. Both types of profile are seen in slit spectrograph data (e.g., D. E. Innes et al. (1997, 2015), and the references therein); however, MOSES observed EEs with more complicated morphologies than either of these two models suggest (J. L. Fox et al. 2010; T. L. Rust 2017). It is unclear whether the differing observations are a function of wavelength and temperature, a result of a limited number of observations, or a consequence of the morphology of the event being difficult to ascertain from slit spectrograph data.

ESIS will observe magnetic reconnection in the context of EEs, by extending the technique pioneered by MOSES to additional TR lines. Explosive events are well suited to sounding rocket observations; a significant portion of their temporal evolution can be captured in  $>150$  s (e.g. the analysis by T. L. Rust (2017)) and they are sufficiently common to provide a statistically meaningful sample in a 5-minute rocket flight (e.g., K. P. Dere et al. (1989, 1991)). In similarity with MOSES, we sought a TR line for ESIS that is bright and well enough isolated from neighboring emission lines so as to be easily distinguished.

### 3.2. Energy Transfer

Tracking the mass and energy flow through the solar atmosphere is a long-standing goal in solar physics. Bulk mass flow is evidenced by Doppler shifts or skewness in spectral lines. However, the observed non-thermal broadening of TR spectral lines may result from a variety of physical processes, including MHD waves (B. De Pontieu et al. 2015, 2007), high-speed evaporative up-flows (e.g. nanoflares, S. Patsourakos & J. A. Klimchuk (2006)), turbulence, and other sources (e.g. J. T. Mariska (1992)). This is a broad topic which ESIS can address in many ways. Here we will focus on a single application; ESIS will search for sources of Alfvén waves in the solar atmosphere by observing Doppler shifts and line broadening as the spectroscopic signature of these waves. The spectroscopic signature of Alfvén waves is most obvious near the limb of the solar disk, since the

motion of the ions is transverse with respect to the propagation direction. Pointing near the limb may make it more difficult to observe distinct EEs, described in Section 3.1, since the average width of TR lines often increases near the limb of the Sun (T. R. Ayres 2021), thus decreasing the signal-to-noise ratio (SNR) of single events.

Alfvén waves in coronal holes are observed to carry an energy flux of  $7 \times 10^5$  erg/cm<sup>2</sup>/s, enough to energize the fast solar wind (M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). The source and frequency spectrum of these waves is unknown. Here, we hypothesize that MHD waves are similarly ubiquitous in quiet Sun and active regions, and play an important role in the energization of the quiescent corona.

The magnitude of non-thermal broadening of optically thin spectral lines is a direct measure of the wave amplitude (D. Banerjee et al. 2009; M. Hahn et al. 2012; M. Hahn & D. W. Savin 2013). We may estimate a lower limit on the non-thermal velocity to be observed as follows. We assume that the magnetic field is constant for small changes in scale height in the TR and that line-of-sight effects are negligible for observations sufficiently far from disk center. Since the solar wind is not accelerated to an appreciable fraction of the Alfvén wave velocity at altitudes  $R \leq 1.15R_\odot$  (S. R. Cranmer & A. A. van Ballegooijen 2005), the wave amplitude,  $v_{nt}$ , depends only weakly on electron density,  $n_e$ , so that  $v_{nt} \propto n_e^{-1/4}$  (M. Hahn & D. W. Savin 2013; T. G. Moran 2001). Assuming pressure balance between the low corona and transition zone, we may infer non-thermal velocities in the TR by scaling according to the temperature drop,  $v_{nt} \propto T^{1/4}$ . The measured non-thermal velocity of  $24 \text{ km s}^{-1}$  for Si VIII (J. G. Doyle et al. 1998) (0.8 MK (T. G. Moran 2003)) near the limb should, neglecting damping, correspond to velocities of at least  $18 \text{ km s}^{-1}$  in the O v (0.25 MK) line. This is larger than the thermal width of  $11 \text{ km s}^{-1}$ .

More recently, A. K. Srivastava et al. (2017) observed torsional Alfvén waves with amplitude  $\sim 20 \text{ km s}^{-1}$  and period  $\sim 30$  s in the chromosphere. Modeling shows that these torsional waves can transfer a significant amount of energy to the corona (T. Kudoh & K. Shibata 1999). The torsional motion is observed as Doppler shifts when viewed from the side. The oscillation period is long enough to be well resolved but short enough to see  $\sim 10$  cycles in a single rocket flight. An ESIS-like instrument is therefore well suited to observations of torsional Alfvén wave propagation over multiple heights in the TR.

By mapping Doppler velocities over a wide field of view in the TR, ESIS can address questions about both

the origin of waves and whether they are able to propagate upward into the corona. Independent of the two propagation modes discussed above, there is a range of possible sources for Alfvén (and other MHD) waves in the solar atmosphere. Three potential scenarios are: (1) Waves originate in the chromosphere or below and propagate through the TR at a spatially uniform intensity; (2) Intense sources are localized in the TR, but fill only a fraction of the surface; and (3) Weak sources are localized in the TR, but cover the surface densely enough to appear like the first case. The resulting non-thermal widths for localized sources are significantly higher than the  $\sim 18 \text{ km s}^{-1}$  lower limit derived above. The concentration of non-thermal energy observed by ESIS will serve as an indicator of source density. Comparison of Doppler maps captured at different temperatures by ESIS will indicate whether a uniform source density originates in the chromosphere or below (scenario 1) or is associated with spatially distributed TR phenomena (scenario 3) such as explosive events, or macrospicules. Comparison with a wider selection of ground- and space-based imagery will allow us to determine whether intense, localized sources (scenario 2) are associated with converging or emerging magnetic bipoles, type II spicules, spicule bushes, or other sources beneath the TR. For these comparisons, we need only to localize, rather than resolve, wave sources. A spatial resolution of  $\sim 2 \text{ Mm}$  is sufficient to localize sources associated with magnetic flux tubes that are rooted in photospheric inter-granular network lanes (e.g. T. E. Berger et al. (1995)).

### 3.3. Science Requirements

ESIS will investigate two science targets: reconnection in explosive events and the transport of mass and energy through the transition region. The latter may take many forms, from MHD waves of various modes to EUV jets or macro-spicules. To fulfill these goals, ESIS will obtain simultaneous intensity, Doppler shift and line width images of the O V 629.732 Å line in the solar transition region (0.25 MK) at rapid cadence. The bright, optically thin O V emission line is well isolated except for the two coronal Mg X lines. These coronal lines can be viewed as contamination or as a bonus; we expect that with the four ESIS projections it is possible to separate the O V emission from that of Mg X. From the important temporal, spatial, and velocity scales referenced in Sections 3.1 and 3.2 we define the instrument requirements in Table 1 that are needed to meet our science goals.

## 4. THE ESIS INSTRUMENT

ESIS is a multi-projection slitless spectrograph that obtains line intensities, Doppler shifts, and widths in a single snapshot over a 2D FOV. Starting from the notional instrument described in Section 2, ESIS has been designed to ensure all of the science requirements set forth in Table 1 are met. The final design parameters are summarized in Table ??.

A schematic diagram of a single ESIS channel is presented in Figure ??a, while the mechanical features of the primary mirror and gratings are detailed in Figures ??b and ??c, respectively.

### 4.1. Pointing System

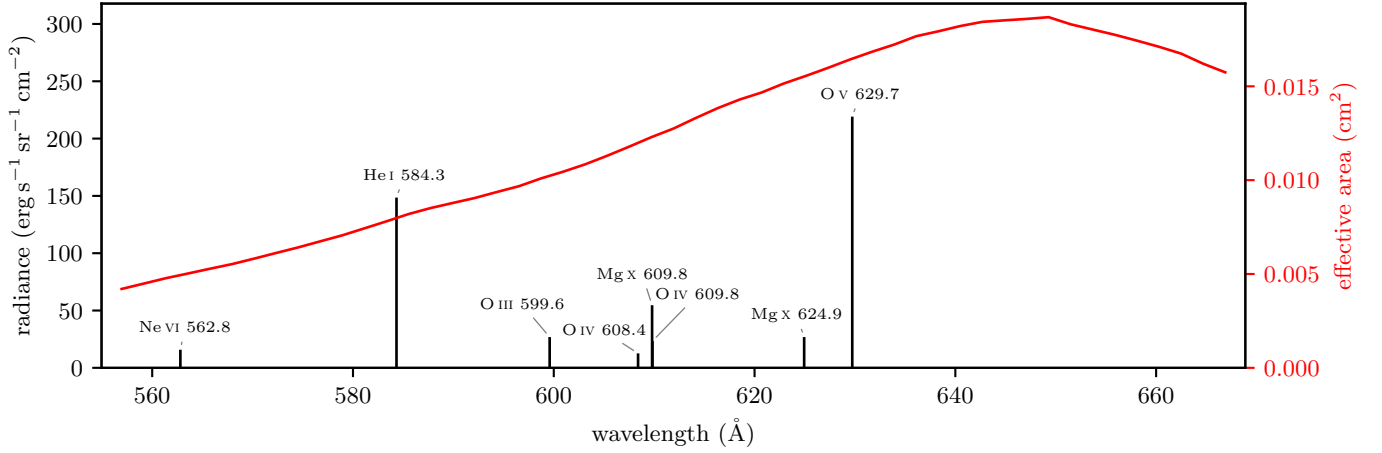
The imaging target was selected prior to launch, the morning of the day of flight. During flight, pointing was maintained by the Solar Pointing Attitude Rocket Control System (SPARCS) (Lockheed Missiles and Space Co. Report 1969). Images from Camera 1 were down-linked and displayed in real time on the SPARCS control system console at intervals of  $\sim 16 \text{ s}$  to verify pointing was maintained during flight.

### 4.2. Mechanical

ESIS and MOSES are mounted on opposite sides of a composite optical table structure originally developed for the Solar Plasma Diagnostics Experiment (SPDE) (M. E. Bruner 1995). The layered carbon fiber structure features a convenient, precisely coplanar array of threaded inserts with precision counterbores. The carbon fiber layup is designed to minimize the longitudinal coefficient of thermal expansion. The optical table is housed in two 0.56 m diameter skin sections, with a total length of 3 m. A ball joint and spindle assembly on one end and flexible metal aperture plate on the other hold the optical table in position inside the skin sections. The kinematic mounting system isolates the optical table from bending or twisting strain of the skins.

### 4.3. Avionics

The ESIS data acquisition and control system (DACS) is based on the designs used for both Chromospheric Lyman-alpha Spectro-polarimeter (CLASP) (R. Kano et al. 2012; K. Kobayashi et al. 2012; N. Narukage et al. 2016) and High-Resolution Coronal Imager (Hi-C) (K. Kobayashi et al. 2014; L. A. Rachmeler et al. 2019). The electronics are a combination of military off-the-shelf (MOTS) hardware and custom designed components. The DACS is a 6-slot, 3U, open VPX PCIe architecture conduction cooled system using an AiTech C873 single board computer. The data system also includes a MOTS PCIe switch card, Marshall Space Flight



**Figure 3.** The eight brightest emission lines in the ESIS passband, and the effective area of a single channel beneath them. The lines are computed from version 11.0.2 of the CHIANTI atomic database, using the J. T. Schmelz et al. (2012) abundances of `sun_coronal_2012_schmelz_ext`, the quiet Sun differential emission measure (DEM) file `quiet_sun.dem`, and  $n_e T = 1 \times 10^{15} \text{ K cm}^{-3}$ .

**Table 1.** ESIS instrument requirements and capabilities. Note that MTF exceeds the Rayleigh criterion of 0.109.

| Parameter          | Requirement           | Science Driver            | Capabilities                                         |
|--------------------|-----------------------|---------------------------|------------------------------------------------------|
| Spectral line      | O v 629.732 Å         | EEs                       | O v, Mg x, He I, Figure 3                            |
| Spectral sampling  | 18 km s <sup>-1</sup> | Broadening from MHD waves | 17.5 km s <sup>-1</sup> pix <sup>-1</sup> , Table ?? |
| Spatial resolution | 2.1'' (1.5 Mm)        | EEs                       | ??, Table ??                                         |
| SNR                | 17.3 (CH)             | MHD waves in CH           | ?? (12 × 10 s exp.), Table ??                        |
| Cadence            | 15 s                  | Torsional waves           | 10 s eff., Section ??                                |
| Observing time     | 150 s                 | EEs                       | 306.8 s, Section ??                                  |
| FOV diameter       | 10'                   | Span QS, AR, and limb     | 11.5', Table ??                                      |

Center (MSFC) parallel interface card, and two MOTS Spacewire cards. A slot for an additional Spacewire card is included to accommodate two more cameras for the next ESIS flight. The C873 has a 2.4 GHz Intel i7 processor with 16 GB of memory. The operating temperature range for the data system is  $-40^\circ\text{C}$  to  $85^\circ\text{C}$ . The operating system for the flight data system is Linux Fedora 23.

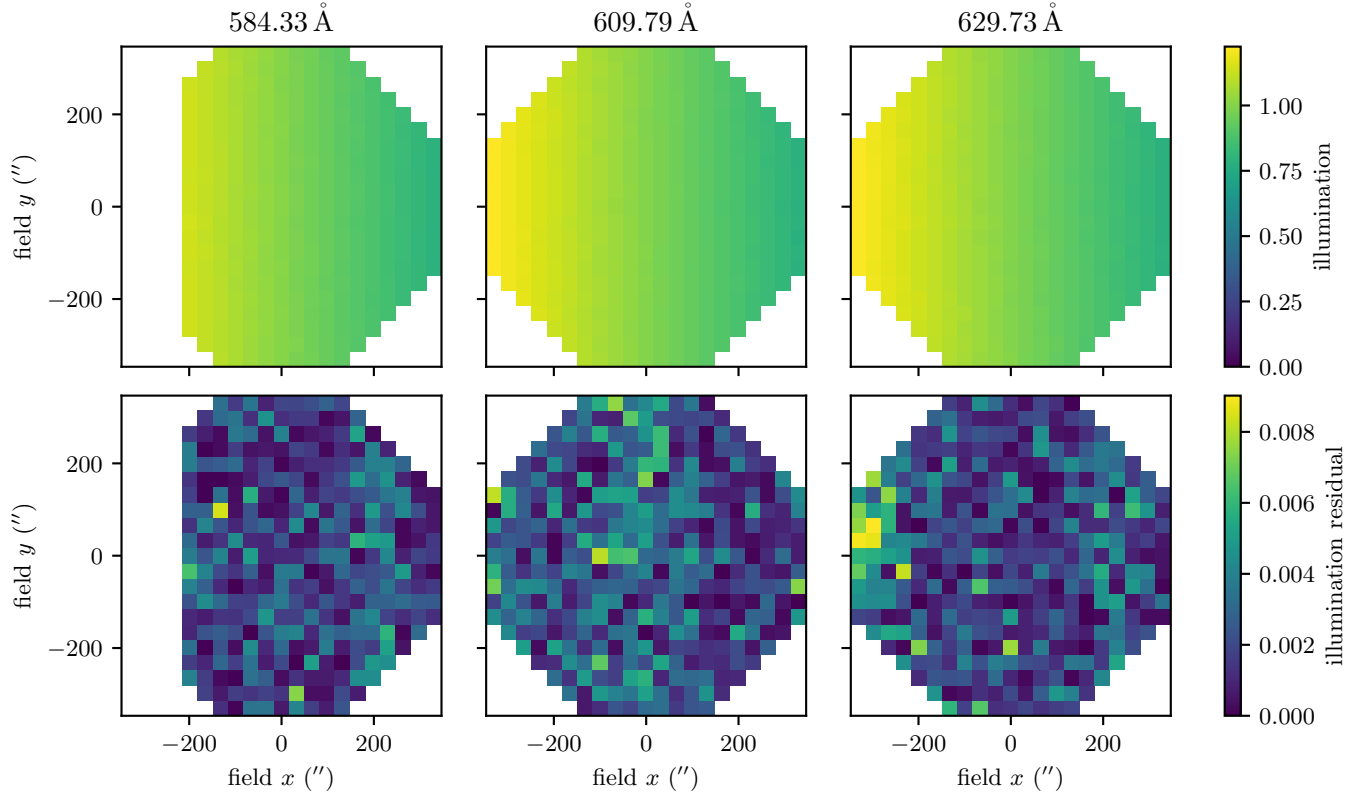
The DACS is responsible for several functions; it controls the ESIS experiment, responds to timers and uplinks, acquires and stores image data from the cameras, downlinks a subset of images through telemetry, and provides experiment health and status. The DACS is housed with the rest of the avionics (power supply, analog signal conditioning system) in a 0.56 m to 0.43 m transition section outside of the experiment section. This relaxes the thermal and cleanliness constraints placed on the avionics. Custom DC/DC converters are used for secondary voltages required by other electronic components. The use of custom designed converters allowed additional ripple filtering for low noise.

#### 4.4. Vignetting

The original design of ESIS had no vignetting thanks to a stop placed at the primary mirror that was designed to perfectly fill the grating with the same amount of light for each point in the FOV. This is the ESIS design that was used for the optimization procedure of the grating parameters described in Section ??, for example. All other results described in the paper use the fully-open system. Before flight, we decided to remove the primary aperture stop to increase the sensitivity of the instrument at the expense of introducing vignetting to the ESIS FOV. This was acceptable since the vignetting was found to be a simple linear field as shown in Figure 4, and could be removed in the post-processing phase.

#### 4.5. Measured Distortion

Every channel of ESIS forms a dispersed image of the same field, and recovering line intensities, Doppler shifts and widths from the four images requires knowing where each point of the sky lands on each detector to a fraction of a pixel, in every exposure. The as-built model of the



**Figure 4.** (Top) The relative illumination of a single ESIS channel as a function of position in the FOV, at each of the three target lines in the passband. The illumination is the fraction of the pupil which is unvignetted, normalized so that its average over the FOV is unity. (Bottom) The residual between that illumination and a linear model of it. Both rows are plotted only where the model was fit: the cells inside the FOV whose light lands on the detector. In the leftmost column, at He I, the FOV is cut off on the left by the edge of the detector.

optics, assembled from the measured figures of the components and their nominal placement, correlates with the flight images at only 0.35 to 0.42; the placement of the optics as flown differs from nominal by enough to move the images by several pixels. We therefore fit the model to the flight itself, and the fit is described in Appendix A. In brief, each channel’s grating angles, ruling spacing, pointing and sensor placement were fit against the exposure at the middle of the flight, using the AIA 304 and 193 Å images of the same field as a proxy for the scene; the windows the field stop cuts into each image were placed on their measured edges, which separates the pointing from the gratings; the four channels were registered against one another on the sky at He I and O v; and the pointing, the drift of the windows and the focus of the primary mirror were followed exposure by exposure. Table 2 gives the correlation of each channel’s model with its exposure after each stage, and on exposures across the flight that the fit never saw.

Figure 5 shows what moved during the flight. The pointing drifted by 6.2'' in yaw and 1.4'' in pitch, and the windows drifted by up to 1.5 pix, which the model reproduces as a translation of the field stop of a few mi-

crons. With those applied, the channels’ skies still slid against one another by 0.23 pix rms, and by 0.51 pix at the first exposure: a translation of each channel’s whole image inside its window, the same at both lines and smooth in time, which nothing after the field stop can produce. Each channel views the Sun through its own sector of the primary mirror, and a change of focus moves its image along its own dispersion while the field stop’s edges stay put; the motion is reproduced, to the precision of the measurement, by a focus that differs from sector to sector and drifts through the flight (Appendix A). The focus of the primary as a whole drifted by 26 μm, and the sectors departed from that mean by up to 18 μm, with the sector of channel 1 moving most; a paraboloid has one focus for every zone, so if this is real it is the figure of the mirror changing unevenly with temperature, by about 100 nm of sag between sectors. The edges the measurement rests on are soft on several sides and the two lines disagree on their motion by up to half the effect, so the data do not exclude that part of the sector pattern is the apparent position of those edges drifting; the model carries the simpler reading, one focus per sector, with that caveat. With it, over the 27

exposures bright enough to measure, the channels agree to 0.11 pix rms at He I and 0.08 pix at O V, and at worst 0.21 pix and 0.24 pix. What the sector focus leaves, 0.08 pix rms in the registration measurement itself, is also fit as an empirical offset of each channel's pointing, of up to  $0.17''$  (0.22 pix), which halves it to 0.04 pix; the model carries these twelve numbers as an option, off by default, since no mechanism stands behind them

(Appendix A). Only the component of a misregistration along a channel's dispersion is a velocity error, and a pixel along the dispersion is  $18.9 \text{ km s}^{-1}$  at He I and  $17.4 \text{ km s}^{-1}$  at O V: projected that way, the registration of the channels contributes  $1.7 \text{ km s}^{-1}$  and  $0.8 \text{ km s}^{-1}$  rms to a measured Doppler shift, and  $3.9 \text{ km s}^{-1}$  at worst, against quiet-Sun velocities of order  $10 \text{ km s}^{-1}$ . The three dark exposures that close the flight are not constrained by the fit and are extrapolated.

## APPENDIX

### A. THE DISTORTION FIT

The distortion of each channel is fit to the flight data as the placement of the optics in the as-built model, rather than as a polynomial over the detector, so that the result has physical units, carries the wavelength dependence the model knows, and can be checked against what was measured on the ground. The fit is reproducible from the `esis` package, whose documentation carries the full report, every table it commits, and the scripts that regenerate them.

*What one exposure determines*—A single exposure of one channel determines eight combinations of that channel's parameters: a singular-value decomposition of the Jacobian of the sky-to-detector mapping with respect to the parameters has eight singular values above 40 pixels of image motion per half-width of the parameter bounds, then a cliff. Nine terms span them: the yaw and pitch of the grating, the ruling spacing, the pitch and yaw of the pointing, and the sensor's distance along the beam and its three angles. The pointing and the grating angles are degenerate in the image of the sky, which only the field stop breaks: its octagonal image, the window of each line on the detector, does not follow the pointing.

*Merit*—Every candidate model is linearized at the three brightest lines of the passband, He I, Mg X and O V, into a polynomial mapping from the sky to the detector and a vignetting function, and the AIA images of the same field at the time of the exposure, 304 Å for He I and O V and 193 Å for Mg X, are resampled through it onto the detector. The merit is the correlation of that model image with the exposure, which is indifferent to the photometric calibration and to the smooth gradients vignetting and effective area leave between the channels. One evaluation takes about a second.

*Stages*—

1. *Capture*. Each channel is fit alone against the exposure at the middle of the flight, a seeded differential evolution over the nine terms followed by a restarted Nelder–Mead polish.
2. *Edges*. An error-function step is fit where each window's predicted outline crosses the rows and columns of every exposure bright enough to show it; the median over the flight places the windows and the departure from it in each exposure is the drift of the windows.
3. *Outline*. Each channel's windows are placed on the median edges through the grating angles and the sensor roll, which move the windows without moving the sky inside them, and the sky terms are re-polished.
4. *Shared*. The pointing is set to one value for the four channels, and each channel's grating and sensor placement is polished again with it fixed.
5. *Internal alignment*. Each channel's exposure is read onto a common sky grid through its own mapping at He I and O V, inside the window that line illuminates, and cross-correlated tile by tile against channel 1; the parameters are moved so that the mappings reproduce the measured shifts. Two lines separate a geometric error, the same shift in both, from a dispersion error.
6. *Focus*. The channels are measured against channel 1 in every exposure. Each channel views the Sun through its own sector of the primary, so a defocus moves its image along its own dispersion while the windows stay put; the

**Table 2.** The correlation of each channel’s model image with its exposure after each stage of the distortion fit, against the reference exposure, and the mean over 6 exposures across the flight that the fit never saw, with each exposure’s pointing applied. The last row is how far the internal alignment leaves each channel’s sky from channel 1’s at the reference exposure, in pixels.

| Stage                    | Channel 0 | Channel 1 | Channel 2 | Channel 3 |
|--------------------------|-----------|-----------|-----------|-----------|
| Capture                  | 0.803     | 0.841     | 0.803     | 0.827     |
| Windows placed           | 0.803     | 0.840     | 0.803     | 0.826     |
| Shared polish            | 0.809     | 0.851     | 0.804     | 0.828     |
| Held out                 | 0.806     | 0.848     | 0.801     | 0.824     |
| Alignment residual [pix] | 0.01      | 0.00      | 0.03      | 0.03      |

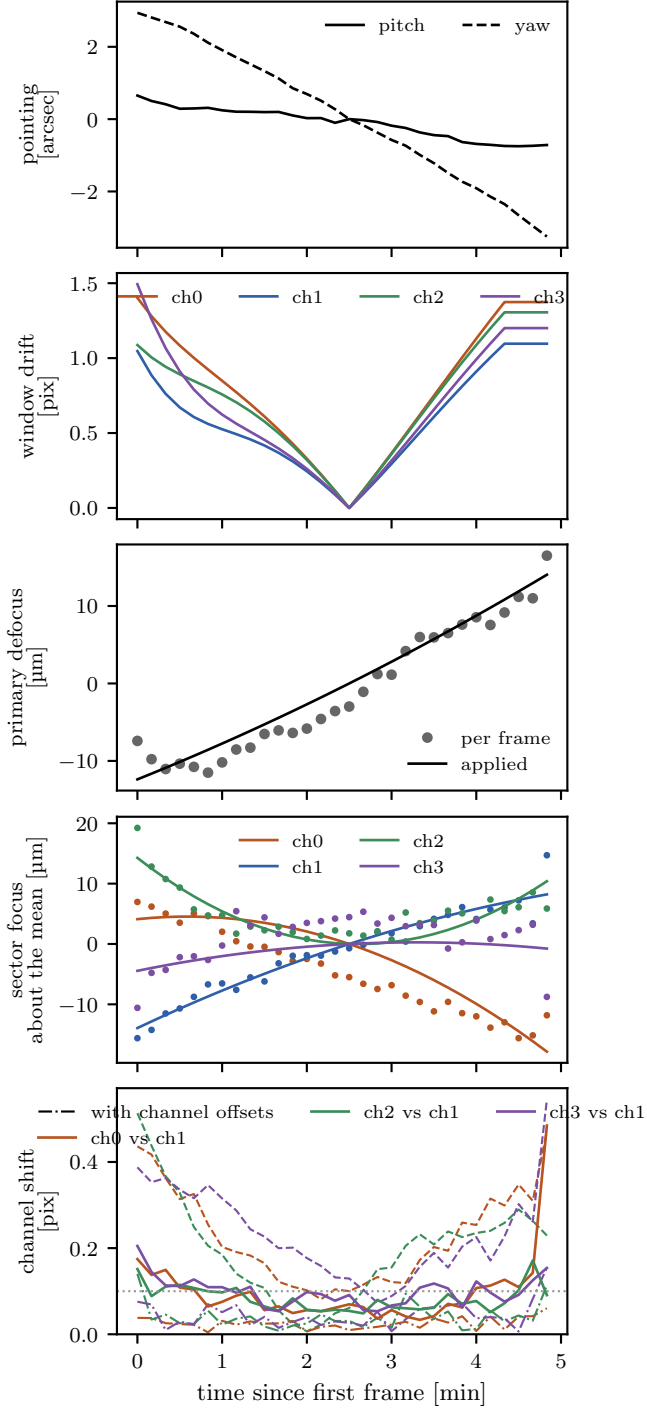
focus of every sector is solved from the shifts through the model, four unknowns from six measured components, and smoothed by a quadratic through the flight. The mean over the sectors is the defocus of the primary as a whole.

7. *Pointing.* The pitch and yaw of every exposure are polished on the mean merit of the four channels, with the roll held and the windows placed by the smoothed drift and the primary defocused by that exposure’s value.

8. *Acceptance.* The model is scored on exposures the fit never saw, and in every exposure the coalignment metric is measured: the length of the median tile shift of each channel against channel 1 at each line, its component along the channel’s dispersion in pixels and in  $\text{km s}^{-1}$ , the scatter of the tiles about it, and the decomposition of the tiles into the linear distortion modes with a test of whether what remains is noise (Figure 6).

The same chain run on a host without a GPU reproduces the committed tables, the correlations to seven figures and the coalignment to 0.01 pixels. A second run from a different random seed, on the chain before the focus of each sector was added, reproduced the correlations to 0.01 and the coalignment to 0.02 pixels; the parameters differ along the null directions of the problem, with the mapping unchanged to about a pixel at the edge of the field.

*The focus of each sector*—With the pointing, the drift and one defocus of the whole primary applied, each channel’s image still translated against the others’ by up to 0.51 pix between the ends of the flight, rigid, the same at both lines and smooth in time. Tracking the interior of each window and its edges separately against the reference exposure, on the detector, the image did not move with the edges, so it was not a motion of a grating or a camera, which carry both; nor a decenter of the field stop or of the primary, which move every window, or every image, by one vector; and an error of plate scale or roll would have to be eight percent or five degrees. A focus that differs from sector to sector moves each channel’s image along its own dispersion by its own amount, three parameters per exposure against six measured components, and it fits: it leaves the floor of the measurement, where the figure modes tried first, astigmatism, coma and trefoil, which force the sectors into a fixed relation, left three times that. Each sector’s focus is smooth in time to the precision of a single exposure, a few microns. Nothing after the field stop can do this, and neither can any rigid motion of the mirror or the stop, so if it is real it is the figure of the primary changing unevenly with temperature, about 100 nm of sag between sectors over the flight, the same scale as the common drift of the focus. But the window edges, which the measurement takes for a rigid outline, are not one. The He I window has no left edge on any detector, where the image runs off the sensor, and two channels lack an edge in the other direction, so several windows are placed along an axis by a single edge. The edges differ in sharpness from 0.7 to over 4 pixels, several sharpen or soften by a pixel during the flight, their profiles are skewed, and the two lines, which share one field stop and one grating, disagree on the motion of the same side by up to 0.3 pixels over half the flight. The focus of a sector is measured along the channel’s dispersion, and the window’s position along the dispersion comes from the two edges perpendicular to it, which are the edges that behaved worst; the crossing of a skewed step moves when its width changes, by a fraction of the change, and that would appear here as a focus of the sector. The disagreement between the lines puts that at up to half the effect. The data therefore support two readings, the mirror bending or the along-dispersion fiducials drifting, and do not decide between them; the model carries the first, the simpler physical account, and the tables record the sectors’ histories so that a thermal model of the mirror, or the next flight’s temperature sensors, can judge it.



**Figure 5.** The distortion fit through the flight. From the top: the pointing of the payload fit to each exposure against the AIA scene; the drift of each channel's windows measured from the edges of the field stop; the defocus of the primary mirror at the field stop, the mean over its sectors, measured from the channels against one another in every exposure and smoothed by a quadratic; the focus of each channel's sector of the primary about that mean, measured per exposure (dots) and as applied (lines); and the shift of each channel's sky against channel 1, dashed with one focus for the whole primary and solid with one per sector, with the 0.1 pix acceptance threshold dotted.

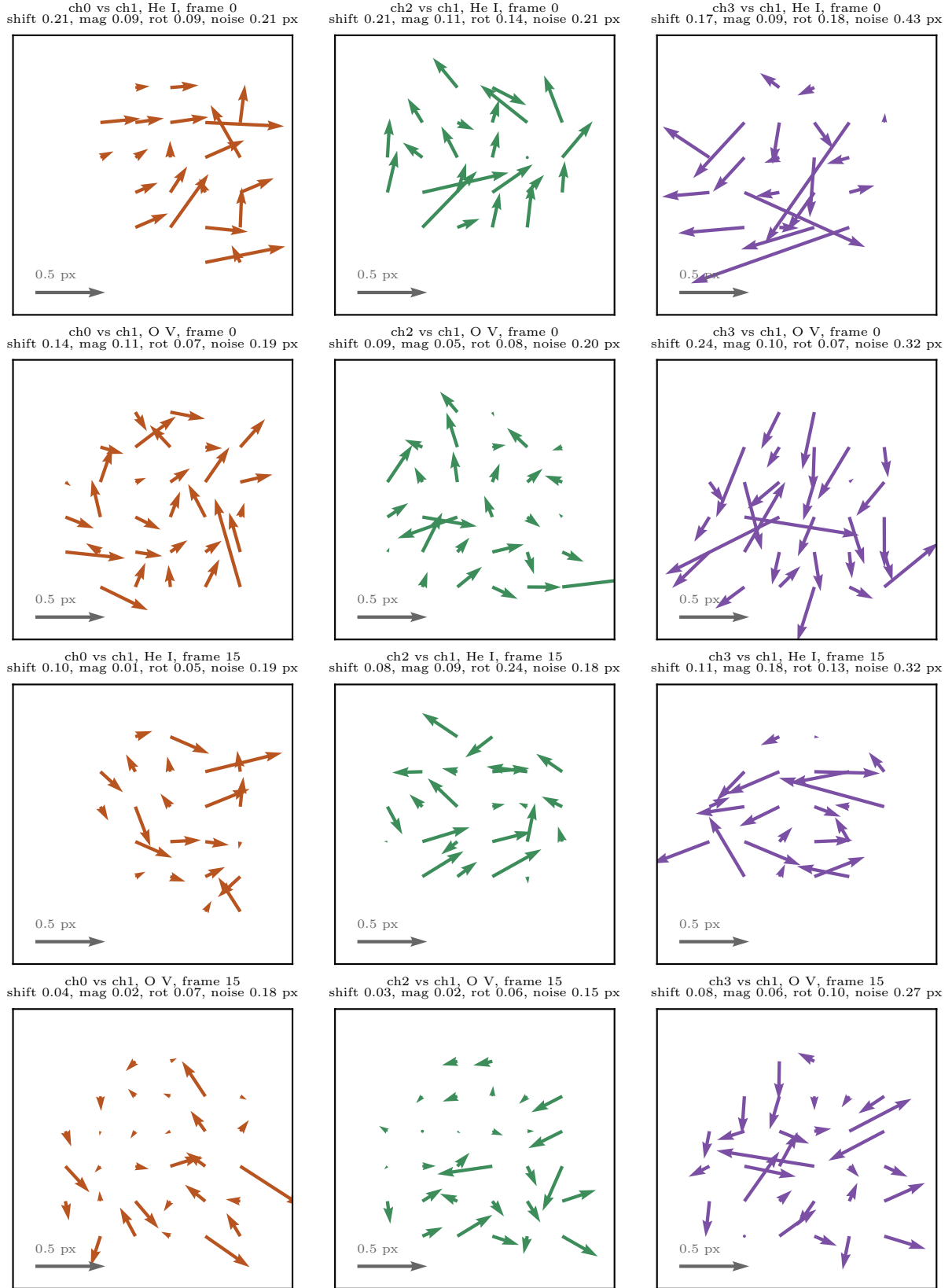
819 *The optional channel offsets*—With the focus of each sector applied, the channels’ skies still differ from channel 1’s by  
 820 0.08 pix rms over the bright exposures, close to the 0.05 pixels a single measurement resolves. The same measurement,  
 821 repeated with the sector focus in place, fits what is left as an offset of each channel’s own pitch and yaw, a quadratic  
 822 in the exposure index that vanishes at the reference exposure and sums to zero over the channels so that the payload’s  
 823 pointing stays where the scene put it: twelve numbers, at most  $0.17''$  or 0.22 pix on the sensor, which leave 0.04 pix  
 824 rms. They are committed beside the fit and the flight model applies them only when asked, since they are empirical  
 825 where the sector focus is a mechanism; an inversion can be run with and without them to see what a few hundredths  
 826 of a pixel of registration do to the result.

827 *Lessons*—The field stop’s edge was the calibration source, and its weak point: a field stop imaged whole, with margin  
 828 on every side of every window, with edges verified sharp on the ground and a deliberate fiducial in each side, would  
 829 have decided the question above and would fix the rotation and the scale of every window as well as its position. An  
 830 undispersed channel sharing the field stop would anchor the sky absolutely to a few hundredths of a pixel, where the  
 831 AIA proxy anchors it to half a pixel, and would separate a common Doppler shift from a change of focus outright. The  
 832 primary’s focus drifted by  $26\text{ }\mu\text{m}$  and the field stop by a few microns, and each moved the images by tenths of a pixel;  
 833 an athermal metering structure between them, or a way to measure both in flight, is worth more to the registration of  
 834 the channels than any improvement of the detectors.

835 *Cost*—The capture takes one to two hours per channel on one GPU, the later stages four hours more, and the chain  
 836 is about six hours of wall time on a cluster; the same chain on the host without a GPU takes about as long, and the  
 837 stages from the focus on fit a workstation with 128 GB of memory.

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**Figure 6.** The coalignment of the channels over the field, after the fit. Each panel shows one channel's sky against channel 1's at one exposure and one line: the field is divided into tiles, each tile of the two channels is cross-correlated, and the shift that best registers them is drawn as an arrow at the tile's place in the field, with a half-pixel arrow for scale. The title of each panel decomposes the arrows into a translation, a magnification and a rotation of the field, and gives what remains once those are removed, which is uncorrelated from tile to tile and is the floor of the measurement. The first exposure of the flight is on top and the reference exposure, against which the fit was made, below.

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